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SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME: Deep Learning **COURSE CODE: MLA04**
SLOT :

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10% **Duration: 90 minutes**
Assessment Tool Weightage: 30% **Mark : 25**

Code of Conduct:
I K.Neeha(192425245) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

K.Neeha
Signature of the student: **Name of the student:K.Neeha**

Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation **(2.5)**	Critical Analysis & Technical Evaluation **(2)**	Future Scope, Presentation & Documentation ***(1)***	**Total (10)**
Weightage	***20 %***	***25%***	***25%***	***20 %***	***10 %***	**100 %**
Q1						
Total						

1. Article Details

Title: DenseNet201Plus: Cost-effective Transfer-Learning Architecture for Rapid Leaf Disease Identification with Attention Mechanisms

Authors: Md Khairul Alam Mazumder, Md Mohsin Kabir, Ashifur Rahman, Md Abdullah-Al-Jubair, M. F. Mridha (PubMed)

Journal: Heliyon

Year of Publication: 2024 (PubMed)

DOI: 10.1016/j.heliyon.2024.e35625 (PubMed)

Publication Link: https://doi.org/10.1016/j.heliyon.2024.e35625

Deep Learning Technique: Transfer Learning, DenseNet201, Attention Mechanism, CNN-Based Classification.

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2. Research Problem

Problem Addressed

Plant leaf diseases significantly reduce agricultural productivity and food security. Early disease detection is necessary for effective treatment and crop protection.

Importance

- Prevents crop losses.
- Improves agricultural yield.
- Supports food security.
- Enables timely disease management.

Limitations of Existing Methods

- Low classification accuracy.
- Poor feature extraction.
- Overfitting on small datasets.
- Limited interpretability.
- High computational cost.

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3. Objectives of the Study

Main Objectives

- Develop an improved DenseNet-based architecture.
- Enhance disease detection accuracy.
- Improve feature extraction through attention mechanisms.
- Increase model robustness and interpretability.

Specific Goals

- Detect plant diseases from leaf images.

- Reuse learned features efficiently.
- Achieve higher accuracy than existing CNN models.

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4. Dataset Description

Datasets Used

1. Banana Leaf Disease Dataset
2. Black Gram Leaf Disease Dataset

Data Type

Image Data (Leaf Images)

Dataset Size

The study uses publicly available datasets from Mendeley Data repositories.

Dataset Split

- Training Set
- Validation Set
- Testing Set

The authors used standard train-validation-test partitioning during experimentation.

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5. Proposed Methodology

Overall Methodology

1. Image Collection
2. Data Preprocessing
3. Data Augmentation
4. Transfer Learning using DenseNet201
5. Attention Module Integration
6. Disease Classification
7. Performance Evaluation

Deep Learning Architecture

DenseNet201Plus consists of:

- DenseNet201 Backbone
- Attention-Based Transition Layers
- Dense Blocks
- Global Average Pooling
- Fully Connected Layer
- Softmax Classifier

Data Preprocessing

- Image Resizing

- Normalization
- Data Augmentation
    - Rotation
    - Flipping
    - Scaling

Training Procedure

- Transfer Learning
- Fine-Tuning
- Cross-Entropy Loss
- Adam Optimizer
- Batch Training

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6. Deep Learning Technique Used

Transfer Learning

The DenseNet201 model was pre-trained and adapted for plant disease classification. Lower layers retained learned features while upper layers were fine-tuned.

DenseNet Feature Reuse

DenseNet connects each layer to all previous layers.

Formula:

[

$x_l=H_l([x_0,x_1,x_2,\ldots,x_{l-1}])$

]

Benefits:

- Feature reuse
- Better gradient flow
- Reduced parameters
- Improved accuracy

RNN / LSTM

Not used in this study.

Bidirectional RNN

Not used.

Encoder–Decoder

Not used.

BPTT

Not used because the architecture is CNN-based rather than recurrent.

7. Model Architecture

Simplified Architecture Diagram

Input Leaf Image
│
▼
Preprocessing
│
▼
Data Augmentation
│
▼
DenseNet201 Backbone
│
▼
Attention Modules
│
▼
Dense Blocks
│
▼
Global Average Pooling
│
▼
Fully Connected Layer
│
▼
Softmax Classifier
│
▼
Disease Prediction

Major Components

Component	**Role**
Input Layer	Receives leaf images
Preprocessing	Cleans and normalizes data
DenseNet201	Feature extraction
Attention Module	Focuses on disease regions
Dense Blocks	Feature reuse
Pooling Layer	Dimensionality reduction
Softmax Layer	Final classification

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8. Performance Evaluation

Evaluation Metrics

- Accuracy

- Precision
- Recall
- AUC (Area Under Curve)

Results

Banana Leaf Dataset

- Accuracy = 90.12%
- Precision = 90.12%
- Recall = 90.12%
- AUC = 97.16%

Black Gram Dataset

- Accuracy = 99.50%
- Precision = 99.50%
- Recall = 99.50%
- AUC = 100%

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9. Results and Discussion

Major Findings

- DenseNet201Plus outperformed traditional CNN architectures.
- Attention mechanisms improved disease localization.
- Feature reuse increased classification accuracy.

Best Performing Model

DenseNet201Plus achieved the best overall performance compared with baseline pre-trained CNN models.

Performance Factors

- Transfer Learning
- Dense Connections
- Attention Mechanisms
- Data Augmentation

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10. Advantages of the Proposed Model

1. High classification accuracy.
2. Efficient feature reuse.
3. Better gradient propagation.
4. Reduced overfitting.
5. Improved interpretability using Grad-CAM++.

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11. Limitations of the Study

Author Mentioned Limitations

- Dataset diversity is limited.
- Performance may vary on unseen crops.

Additional Limitations

- High computational requirements.
- Training complexity.
- Large memory consumption.
- Generalization to real-world field conditions may require further testing.

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12. Critical Analysis

Is DenseNet Appropriate?

Yes. DenseNet is highly suitable because plant disease images require fine-grained feature extraction.

Is Dataset Adequate?

The datasets are sufficient for experimentation but larger multi-crop datasets would improve generalization.

Are Metrics Appropriate?

Yes. Accuracy, Precision, Recall, and AUC provide comprehensive evaluation.

Are Results Convincing?

Yes. The reported performance is significantly better than many baseline CNN architectures.

Could Another Technique Perform Better?

Vision Transformers (ViT) or EfficientNet-based transfer learning models may achieve competitive results on larger datasets.

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13. Future Scope

Possible improvements include:

- Multi-crop disease detection.
- Real-time mobile deployment.
- Integration with IoT agriculture systems.
- Vision Transformer integration.
- Larger and more diverse datasets.

- Hybrid DenseNet–Transformer architectures.

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14. Conclusion

This research addresses the challenge of accurate plant disease detection using deep learning. The authors proposed DenseNet201Plus, an enhanced transfer-learning architecture incorporating attention mechanisms and dense feature reuse. The model achieved up to 99.5% accuracy on plant disease datasets and outperformed existing CNN approaches. The study demonstrates the effectiveness of transfer learning and DenseNet architectures for agricultural disease diagnosis. Overall, the article presents a well-designed, practical, and high-performing deep learning solution for intelligent agriculture.

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15. References

1. Mazumder, M. K. A., Kabir, M. M., Rahman, A., Abdullah-Al-Jubair, M., & Mridha, M. F. (2024). *DenseNet201Plus: Cost-effective transfer-learning architecture for rapid leaf disease identification with attention mechanisms*. Heliyon, 10(15), e35625. DOI: 10.1016/j.heliyon.2024.e35625. (PubMed)
2. DenseNet201Plus Article (Open Access):

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context

Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis